

System Architecture & Solution Overview

Multimodal RAG Engine — Apple Inc. Form 10-Q (Q3 2022)

Selected Core Strategy: Single-unit financial table chunking + Hybrid Retrieval (Chroma Dense vector search + BM25Okapi sparse keyword search combined via Reciprocal Rank Fusion, RRF k=60) + Intent-based Query Routing + Cross-Encoder Reranking (ms-marco-MiniLM-L-6-v2) + Safe AST Calculator Tool.

1. Approach & Methodology

- Document Parsing & Normalization:** PyMuPDF extracts text blocks and page hierarchy; pdfplumber extracts financial tables. Negative numbers in parentheses (e.g., (1,234)) are converted to standard negatives -1234, and multi-tier headers are flattened into clean column descriptions.
- Single-Unit Table Chunking:** Text sections are split at 500-token boundaries with 50-token overlap. Financial tables are kept as single atomic markdown units so numbers, rows, and headers are never split across chunks.
- Dual Indexing & Hybrid Retrieval:** Dense semantic vectors (BAAI/bge-base-en-v1.5 in Chroma DB) capture intent, while sparse keyword indexes (rank_bm25) catch exact numbers and financial terms. Results are merged using Reciprocal Rank Fusion (RRF).
- Intent Routing & Reranking:** A lightweight router boosts relevant chunk types (table, text, figure) based on query keywords. Top candidates are then reranked with a Cross-Encoder model to ensure high contextual precision.
- Grounded Generation & Calculator Tool:** Answers must cite explicit pages [p.X, Title] and state units (e.g., millions, \$). Math expressions (YoY %, margins) are evaluated via Python AST to guarantee exact arithmetic.

2. Key Design Decisions & Technical Reasoning

Design Decision	Reasoning & Business Impact
Single-Unit Table Chunking	Splitting tables by token count breaks row alignment and column headers. Keeping each table intact as a markdown unit preserves structural integrity.
Hybrid RRF (Dense + BM25)	Dense embeddings often fail on exact numeric queries or specific page items. BM25 guarantees keyword matches for precise financial data retrieval.
AST Calculator Tool	LLMs frequently miscalculate YoY growth percentages or margins. Parsing expressions like CALCULATE: ((1960-1950)/1950)*100 ensures accurate arithmetic.
Strict Abstention & Guardrails	Out-of-scope questions or casual greetings (hi) previously risked hallucinating financial numbers. Adding greeting detection and refusal logic improves reliability.

3. Assumptions, Challenges & Future Improvements

- Scope Assumption:** System is built and benchmarked specifically for Apple Inc.'s 28-page Q3 2022 Form 10-Q filing. The document is heavily text- and table-centric with minimal graphical figures (cover logo).
- Extracted Table Challenges:** Multi-column financial layouts with indented sub-line items were resolved by concatenating parent section titles into chunk metadata path strings.
- Non-Financial Query Handling:** Handled non-question inputs (hi, hello, who are you) gracefully by returning explicit abstention rather than retrieving arbitrary context.
- Future Improvements:** Extend dual-index layer to support multi-period 10-K/10-Q filing trends, introduce Table-to-SQL execution for multi-table aggregations, and add vision LLM extraction for chart-heavy filings.